Piecewise Linear Regression

Optimization problem:

$$\min_{x} \quad f(x) = \sum_{i=1}^{n} |a_i x - b_i|$$

with data pairs $\{(a_i, b_i)\}_{i=1}^{n}$.

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Equivalent linear programming:

$$\min_{x, t_1, \ldots, t_n} \quad \sum_{i=1}^{n} t_i$$

$$\text{s.t.} \quad -t_i \leq a_i x - b_i \leq t_i, \quad \forall i$$

Piecewise Linear Regression

How to minimize $\sum_{i=1}^{n} |a_i x - b_i|$ without using linear programming?

Piecewise Linear Regression

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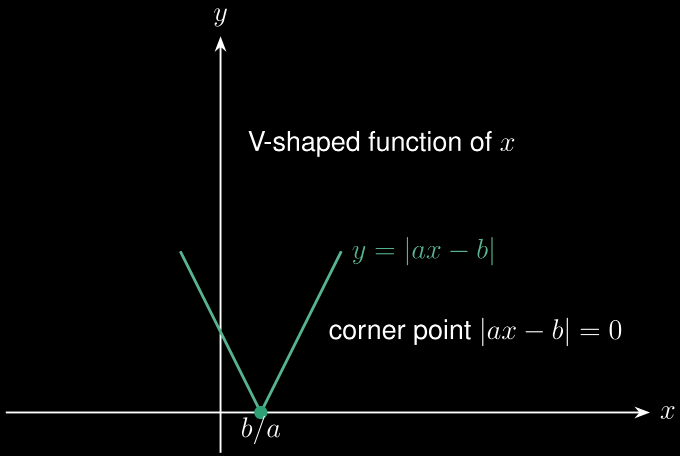

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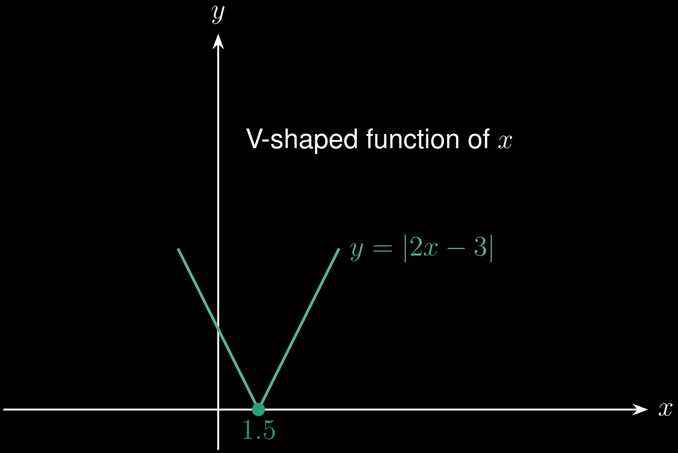

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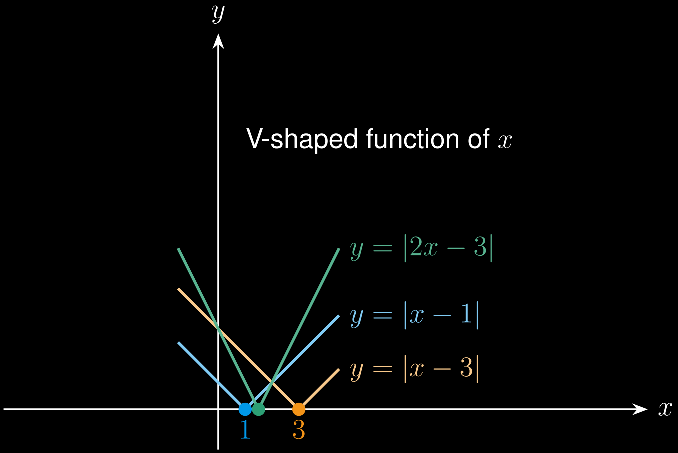

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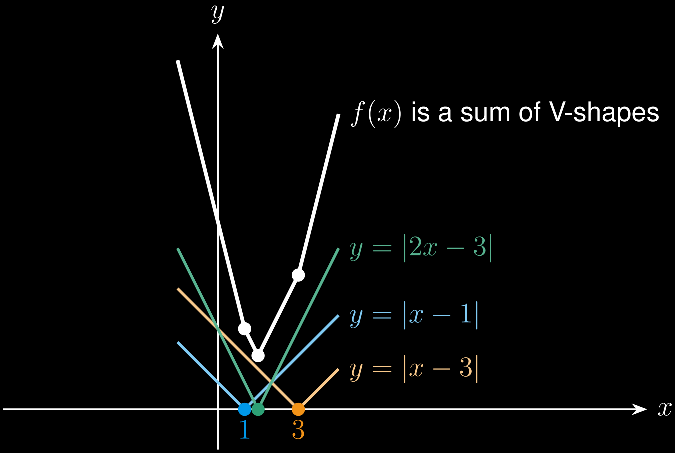

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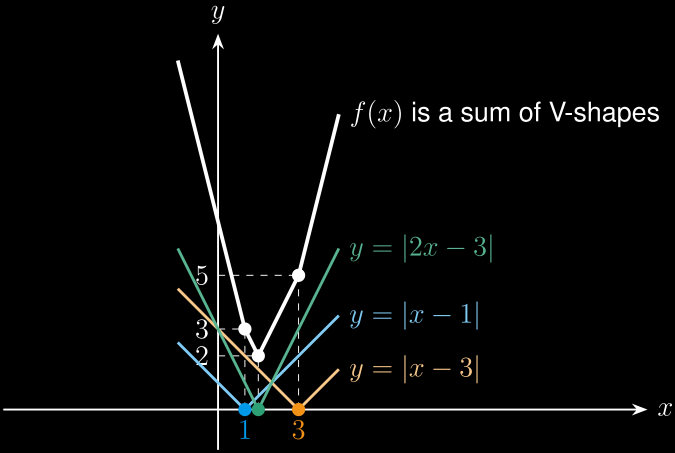

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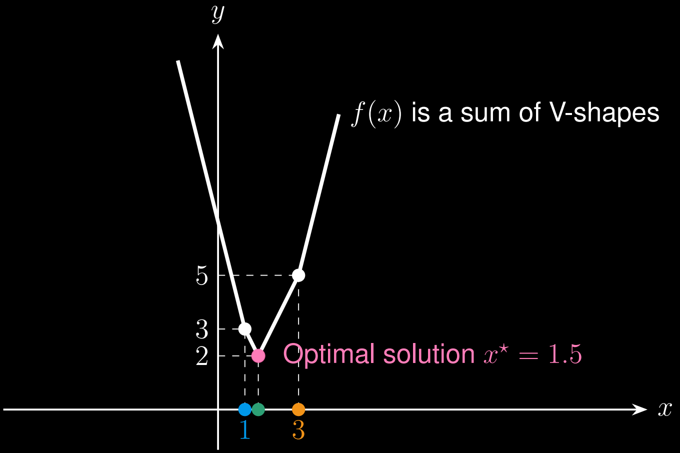
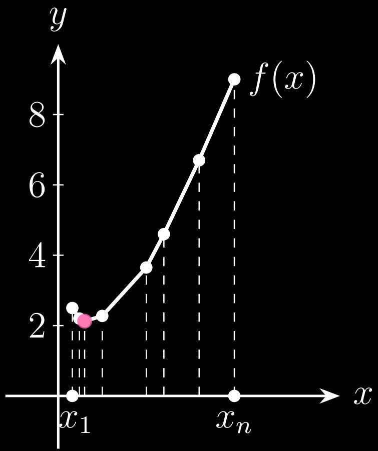


Piecewise Linear Regression without Using Linear Programming

$$\min_{x} \quad f(x) = \sum_{i=1}^{n} |a_i x - b_i|$$

a_i	b_i	$x_i = b_i/a_i$
0.4	0.5	1.25
0.5	0.2	0.4
0.1	0.3	3
0.2	0.5	2.5
0.4	0.3	0.75
0.1	0.5	5
0.1	0.4	4
0.2	0.5	2.5
0.5	0.3	0.6

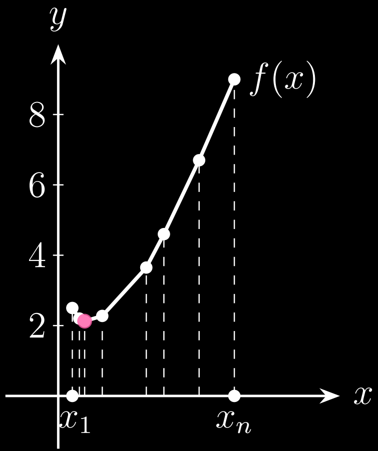


Piecewise Linear Regression without Using Linear Programming

$$\min_{x} \quad f(x) = \sum_{i=1}^{n} |a_i x - b_i|$$

Sort values $x_1 \leq x_2 \leq \ldots \leq x_n$:

$x_i \in \{0.4,\ 0.6,\ 0.75,\ 1.25,\ 2.5,\ 2.5,\ 3,\ 4,\ 5\}$



Piecewise Linear Regression without Using Linear Programming

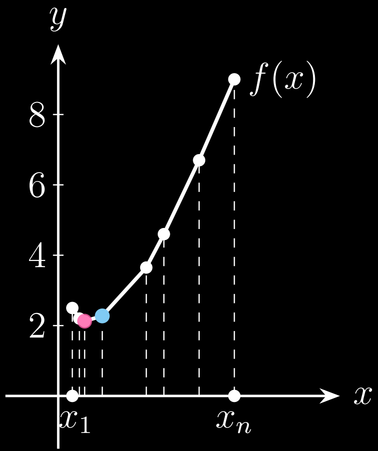
$$\min_{x} \quad f(x) = \sum_{i=1}^{n} |a_i x - b_i|$$

Compute $k = \lfloor \frac{n}{2} \rfloor = 4$:

$$x_i \in \{0.4,\ 0.6,\ 0.75,\ \underbrace{1.25}_{x_4},\ 2.5,\ 2.5,\ 3,\ 4,\ 5\}$$

- Positive slope on $[x_4, x_5]$
- Positive slope on $[x_3, x_4]$

$f(x_4)$ is not the minimum value.



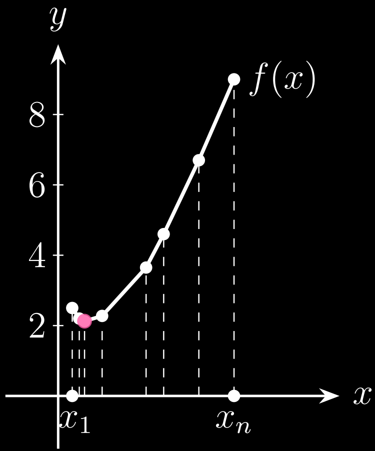
Piecewise Linear Regression without Using Linear Programming

$$\min_{x} \quad f(x) = \sum_{i=1}^{n} |a_i x - b_i|$$

$$x_i \in \{0.4,\ 0.6,\ \underbrace{0.75}_{x_3},\ 1.25,\ 2.5,\ 2.5,\ 3,\ 4,\ 5\}$$

- Positive slope on $[x_3, x_4]$
- Negative slope on $[x_2, x_3]$

$f(x_3)$ is the minimum value.



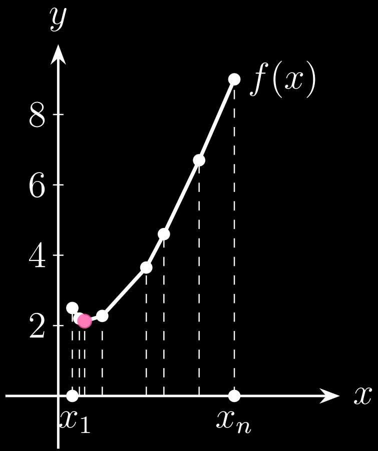
Piecewise Linear Regression without Using Linear Programming

$$\min_{x} \quad f(x) = \sum_{i=1}^{n} |a_i x - b_i|$$

Algorithm:

① Compute all points $x_i = b_i / a_i$;

② Sort values $x_1 \leq x_2 \leq \ldots \leq x_n$;



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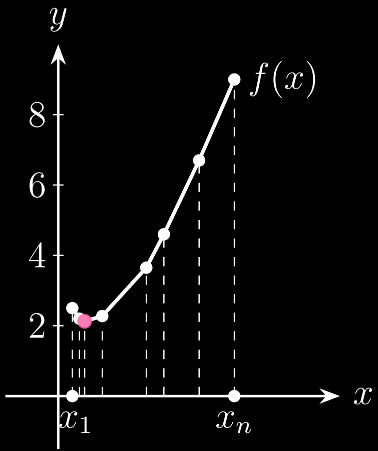
② Sort values $x_1 \leq x_2 \leq \ldots \leq x_n$;

③ Let $k = \lfloor \frac{n}{2} \rfloor$ and compute the slope

$$s = \frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k}$$



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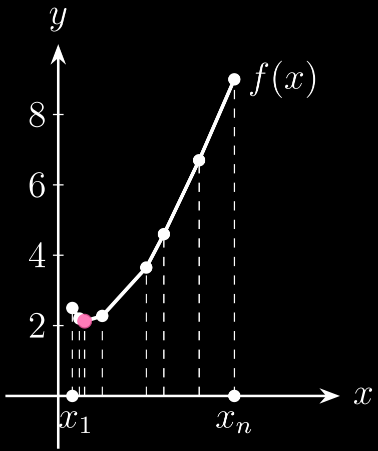
$$s = \frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k}$$

④ Find some x_ℓ such that the slope is negative on $[x_{\ell-1}, x_\ell]$ but positive on $[x_\ell, x_{\ell+1}]$.



Thanks for your attention!

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